

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO4: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	20% of CIA	Total Marks:	100
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
Student Name:	Manoj M	Reg. No.:	192511060
Section / Batch:	CSE	Date:	22.08.2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues.
- Provide a thorough analysis with validated results and performance evaluation.

Question Type	Focus Area	Questions
Industry Problem Solving Task	Focus on Solving optimization problems using dynamic programming and greedy strategies	Q1

SECTION A – Industry Problem Solving Task

Smart Agriculture Irrigation Optimization (Greedy + Scheduling)100 marks

An agricultural system must optimize irrigation schedules to conserve water while maximizing crop yield. Analyze how scheduling algorithms can be applied. Identify constraints such as weather conditions and soil moisture. Design a solution using greedy approaches. Discuss real-time adaptability. Evaluate efficiency and scalability. Interpret results in sustainable agriculture systems.

Code of Conduct

I _____ Manoj (192511060)____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION**Industry Problem Solving Task**

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%				
Application of Engineering Concepts	25%				
Solution Design	25%				
Use of Tools	15%				
Analysis	10%				
Professionalism	5%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name: _____	Name: _____	Name: _____
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Algorithm: Smart Agriculture Irrigation Optimization

1. **Initialize inputs** such as soil moisture, crop water requirement, weather forecast, available water, and irrigation time slots.
2. **Check constraints:** skip or reduce irrigation if rainfall is expected, soil moisture is sufficient, or water availability is low.
3. **Calculate priority** for each field using soil dryness, crop water requirement, growth stage, and weather conditions.
4. **Apply Greedy Scheduling:** select the highest-priority field first and allocate the minimum required water within the available time and water limit.
5. **Repeat** until all possible fields are scheduled or the water/time limit is reached.
6. **Adapt in real time** by continuously updating soil-moisture and weather data and rescheduling irrigation when conditions change.
7. **Evaluate performance** using water saved, crop coverage, irrigation efficiency, and computational time

Code:



SIMATS Online Programming Lab
DAA PROGRAMMING

Manoj M
192511060RC

1/1 answered

Your Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Field {
5     int id;
6     int water_needed;
7     int yield_benefit;
8     double ratio;
9 };
10
11 int compare(const void *a, const void *b) {
12     struct Field *f1 = (struct Field *)a;
13     struct Field *f2 = (struct Field *)b;
14     if (f2->ratio > f1->ratio) return 1;
15     if (f2->ratio < f1->ratio) return -1;
16     return 0;
17 }
18
19 int main() {
20     int n, max_water;
21
22     // Reads input: number of fields and total water available
23     if (scanf("%d %d", &n, &max_water) == 2 && n > 0) {
24         struct Field fields[100];
25         for (int i = 0; i < n; i++) {
26             fields[i].id = i + 1;
27             scanf("%d %d", &fields[i].water_needed, &fields[i].yield_benefit);
28             fields[i].ratio = (double)fields[i].yield_benefit / fields[i].water_needed;
29         }
30         qsort(fields, n, sizeof(struct Field), compare);
31         int water_used = 0;
32         int total_yield = 0;
33         for (int i = 0; i < n; i++) {
34             if (water_used + fields[i].water_needed > max_water) break;
35             water_used += fields[i].water_needed;
36             total_yield += fields[i].yield_benefit;
37         }
38         printf("Scheduled Fields: ");
39         for (int i = 0; i < n; i++) {
40             printf("%d ", fields[i].id);
41             if (i % 10 == 9) printf("\n");
42         }
43         printf("Total Yield: %d\n", total_yield);
44         printf("Water Used: %d\n", water_used);
45     }
46 }
```

Input

```
5 60
20 40
10 30
30 45
```

Output

```
Scheduled Fields: Field 2, Field 5,
Field 1
Total Yield: 120
Water Used: 55
```